

MANOHAR PATTANAYAK

AI Solution Security Architect

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PROFESSIONAL SUMMARY

AI Solution Security Architect with nearly 3 years of experience designing secure architectures for AI-integrated, cloud-native, and enterprise applications across BFSI, Fintech, Insurance, and Healthcare. Specializes in AI security architecture, threat modeling, application security testing, and cloud security across AWS, Azure, and GCP, with a proven record of reducing risk exposure and achieving 100% audit and remediation compliance.

CORE COMPETENCIES

AI & Architecture Security: AI Security Architecture, LLM Security, Prompt Injection Defense, Model Governance, Threat Modeling, Secure SDLC, API Security

Application Security: SAST, DAST, OWASP Top 10, Penetration Testing, AppSec Testing, Security Architecture Reviews

Cloud Security: AWS, Azure, GCP, IAM, Cloud Compliance, Security Posture Management, Encryption

GRC & Compliance: ISO 27001, ISO 42001, SOC 2 Type II, PCI DSS, RBI Framework, ITGC

Programming & Tools: Python, Java, SQL, Bash/Linux Scripting

PROFESSIONAL EXPERIENCE

Consultant - AI & Solution Security Architect

March 2026 - Present

NuSummit Cybersecurity, Mumbai, India

- Conducted network architecture and firewall security reviews (segmentation, VPNs, IDS/IPS), identifying critical security gaps and reducing misconfigurations by 30% across client environments.
- Performed AI security architecture reviews for AI-integrated products, identifying prompt injection, model governance gaps, and LLM data privacy risks, reducing AI risk exposure by 35% across 5+ systems.
- Assessed cloud security architecture across AWS, Azure, and GCP, uncovering IAM misconfigurations and encryption gaps and achieving 100% remediation compliance for 8+ critical applications.
- Performed threat modeling for cloud infrastructure and LLM deployments, delivering prioritized remediation roadmaps that reduced risk exposure by 40% across critical workloads.

Senior Security Analyst - Application Security Architect

August 2023 - March 2026

Deloitte Touche Tohmatsu India LLP, Mumbai, India

- Designed and reviewed application security architectures for 15+ enterprise clients across BFSI, Fintech, and Healthcare, reducing architectural risk exposure by 40%.
- Executed application security testing (SAST, DAST, API security) across 20+ web and mobile applications, uncovering OWASP Top 10 vulnerabilities and reducing critical findings by 45%.
- Performed AI security architecture reviews for AI/ML solutions, identifying model poisoning and adversarial input risks, delivering remediation roadmaps that reduced AI risk exposure by 35%.
- Executed compliance audits aligned with ISO 27001, PCI DSS, SOC 2 Type II, and RBI guidelines, achieving 100% audit clearance for all BFSI clients.

Security Analyst

December 2022 - May 2023

SDI, Bhubaneswar, India

- Performed application and network security testing across web, mobile, and client-server applications, uncovering 50+ vulnerabilities and reducing network risk exposure by 30%.
- Collaborated with development teams to validate security fixes, achieving zero re-occurrence of identified vulnerabilities across all retested systems.

KEY PROJECTS

Security Audit - Payment Gateway System

- Conducted penetration testing across 100+ endpoints, identifying and mitigating 15+ high-severity vulnerabilities and reducing overall attack surface by 35% with zero re-occurrence during retesting.

ATM Security Audit - Logical & Physical Controls

- Assessed security controls for 50+ ATMs, achieving 100% RBI compliance and reducing fraud risk exposure by 20% through targeted mitigation recommendations.

CERTIFICATIONS

ISO 42001:2023 AI Management System Lead Auditor | ISO 27001:2022 Lead Auditor | CEH Practical | Certified AppSec Practitioner (CAP)

EDUCATION

B.Tech in Electronics & Communication Engineering

2019 - 2023

Silicon Institute of Technology, Bhubaneswar, India | CGPA: 8.2 / 10